POWER ENTRY, LOGIC SUPPLY, GROUNDS, DECOUPLING

R1 10R
C1 220u 25V
C2 100n
R2 10R
C3 220u 25V
C4 100n
R3 10R
C5 47u 25V
R4 10R
C6 47u 25V
NT1 AGND–LGND star
CN4 Ground spade 6.3mm
R5 10R GND cancel
U1C OPA1612
C20 100n
C21 100n
U2C OPA1612
C22 100n
C23 100n
U3C OPA1612
C24 100n
C25 100n
U5C OPA1612
C29 100n
C30 100n
U6C OPA1612
C31 100n
C32 100n
U7C OPA1612
C33 100n
C34 100n
U8C OPA1612
C35 100n
C36 100n
U12C OPA1612
C63 100n
C64 100n
U13C OPA1612
C65 100n
C66 100n
U14C OPA1612
C67 100n
C68 100n
U4C LM393
C26 100n
#FLG01 PWR_FLAG
#FLG02 PWR_FLAG
#FLG03 PWR_FLAG
#FLG04 PWR_FLAG
#FLG05 PWR_FLAG
#FLG06 PWR_FLAG

MOTHERLOOM (Delta pinout; one header, crimped socket on a daisy-chained ribbon)



LINE INPUT — ground-compensated receiver (inverting), Ø switch SW15: latched = phase reversed

CN3 Jack board link 2x5
R10 1M
SW15A Phase
SW15B Phase
R6 10K 0.1%
R7 10K 0.1%
C7 33p COG
R8 10K 0.1%
R9 10K 0.1%
U1A OPA1612

GAIN −15..+15 dB, 0 dB at the detent (inverting; pot pin 3 = CW = input end)

R45 2K2 1%
RV9 Gain 10K lin CD
R46 2K2 1%
C37 22p COG
U1B OPA1612
C38 47u NP 25V
R47 47K

HPF unity Sallen-Key Butterworth, −3 dB 87 Hz SW11: latched = HPF in (unlatched shorts the caps)

C39 47n COG
C40 47n COG
R48 27K
R49 56K
U5A OPA1612
SW11A HPF
SW11B HPF

INSERT POSITION JUMPERS 1–2 = insert post-EQ (default); 2–3 on all three = insert pre-EQ

JP6 EQ input
JP5 Insert send source
JP7 PFL source

EQ Delta ED3344A/ED3151 values: HF shelf (U5B), HI-MID (U6A), LO-MID (U6B), LF (U7A); CW = boost

R50 33K
R51 33K
C41 33p COG
U5B OPA1612
RV10 HF 10K lin CD
C42 47n COG
R52 5K6
C43 47n COG
R53 6K8
R54 22K
R55 22K
U6A OPA1612
RV11 HI-MID gain 10K lin CD
R58 22K
R59 22K
U6B OPA1612
RV13 LO-MID gain 10K lin CD

EQ MID TUNING KHN band-pass per mid band: summer (10K x3, BP via 2K2/10K), two integrators, A10K dual

R81 10K 1%
R82 10K 1%
R83 10K 1%
R84 2K2 1%
R85 10K 1%
U12A OPA1612
RV12 HI-MID freq 10K A dual
R86 430R 1%
R87 430R 1%
R88 3K 1%
R89 3K 1%
R56 4K7 1%
R57 4K7 1%
C44 2n2 COG
C45 47n COG
U12B OPA1612
U13A OPA1612
R90 10K 1%
R91 10K 1%
R92 10K 1%
R93 2K2 1%
R94 10K 1%
U13B OPA1612
RV14 LO-MID freq 10K A dual
R95 430R 1%
R96 430R 1%
R97 3K 1%
R98 3K 1%
R60 4K7 1%
R61 4K7 1%
C46 22n COG
C47 22n COG
U14A OPA1612
U14B OPA1612
R66 22K
R67 22K
U7A OPA1612
R62 3K3
R63 3K3
RV15 LF 10K lin CD
C48 150n film
R64 1K5
C49 22u film
C50 100n COG
R65 5K1

EQ IN SW12: latched = EQ in (unlatched = bypass from the EQ input); green EQ LED

SW12A EQ
SW12B EQ
R69 4K7
D5 EQ green
C51 22p NP 25V
R68 47K

INSERT (send = tip, return = ring, ring normalled to send) + return buffer

R11 75R
C10 47u NP 25V
R12 47K
U7B OPA1612
C11 220u NP 25V
R13 47K

OVERLOAD DETECTOR (−+18 dBu)

R39 1K
D1 1N4148
C12 1u
R14 470K
R40 10K
R41 10K
U4A LM393
R15 1K
D2 OL red
U4B LM393

PFL (pre-fade) SW3: A–B = ON

R32 22K
SW3A PFL
SW3B PFL
R33 100K
R34 4K7
D3 PFL red

ON / SOFT MUTE (2x J112 shunt, ~20 ms ramp) SW4: latched = channel ON, green LED

R16 4K7
Q1 J112
R17 4K7
Q2 J112
U3B OPA1612
SW4A ON
R42 220K
C28 100n
SW4B ON
R43 4K7
D4 ON green

FADER + POST-AMP (+10 dB: omit R38 for unity) JP1 = VCA bypass

RV1 Fader 10K log 100mm
C13 2.2u
JP1 VCA bypass
R18 100K
J3 mono 3.5 mm socket
J4 VCA control in
R44 100K
U2A OPA1612
R19 4K7
C14 100p COG
R38 2K2
C15 220u NP 25V
R35 47K

AUX SENDS PRE switch per pair (latched = pre); jumpers 1–2 = pre-fader, 2–3 = pre-EQ. SW13 DIR: aux 6 pot → direct out

JP2 Aux1/2 pre source
SW8A PRE 1–2
SW8B PRE 1–2
JP3 Aux3/4 pre source
SW9A PRE 3–4
SW9B PRE 3–4
JP4 Aux5/6 pre source
SW10A PRE 5–6
SW10B PRE 5–6
RV2 Aux1 10K log
R20 22K
RV3 Aux2 10K log
R21 22K
RV4 Aux3 10K log
R22 22K
RV5 Aux4 10K log
R23 22K
RV6 Aux5 10K log
R24 22K
RV7 Aux6 10K log
R25 22K
SW13A DIR
SW13B DIR

DIRECT OUT (level = aux 6 pot when DIR is latched; impedance-balanced TRS)

R70 100K
U8A OPA1612
R71 75R
C52 47u NP 25V
R72 47K
R73 75R
U8B OPA1612

PAN + PAN BUFFERS (AC-coupled outputs)

RV8 Pan 10K dual
U2B OPA1612
U3A OPA1612
C16 220u NP 25V
R36 47K
C17 220u NP 25V
R37 47K

PAN SWITCH SW14: latched = subgroups panned; unlatched = subgroups get POST, unpanned (MIX always panned)

SW14A PAN
SW14B PAN

ROUTING MATRIX 4K7 summing; DG413 at the bus end: assigned = bus, off = AGND. PVA2: pole A logic, pole B LED

R80 4K7
D9 BZX84C5V1
C62 1u
#FLG07 PWR_FLAG
R26 4K7 1%
R27 4K7 1%
U9A DG413
U9D DG413
U9B DG413
U9C DG413
U9E DG413
C53 100n
C54 100n
C55 100n
SW5A Assign L–R
SW5B Assign L–R
R74 100K
R75 4K7
D6 L–R yellow
R28 4K7 1%
R29 4K7 1%
U10A DG413
U10D DG413
U10B DG413
U10C DG413
U9F DG413
C56 100n
C57 100n
C58 100n
SW6A Assign 1–2
SW6B Assign 1–2
R76 100K
R77 4K7
D7 1–2 yellow
R30 4K7 1%
R31 4K7 1%
U11A DG413
U11D DG413
U11B DG413
U11C DG413
U9G DG413
C59 100n
C60 100n
C61 100n
SW7A Assign 3–4
SW7B Assign 3–4
R78 100K
R79 4K7
D8 3–4 yellow

Nets joined by labels; tidy wiring in Eeschema as needed
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Title: Channel strip — goa console (Delta Theatre input without the mic stag
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